

3.1.GENERATION OF PHASE MODULATED (PM) SIGNALS AND CALCULATION OF PHASE DEVIATION USING MATLAB SIMULATION

AIM : To generate and study generation of phase modulated (pm) signals and calculation of phase deviation using MATLAB .

SOFTWARE REQUIRED : MATLAB

Objective:

i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.

ii)

CODE :

clc;

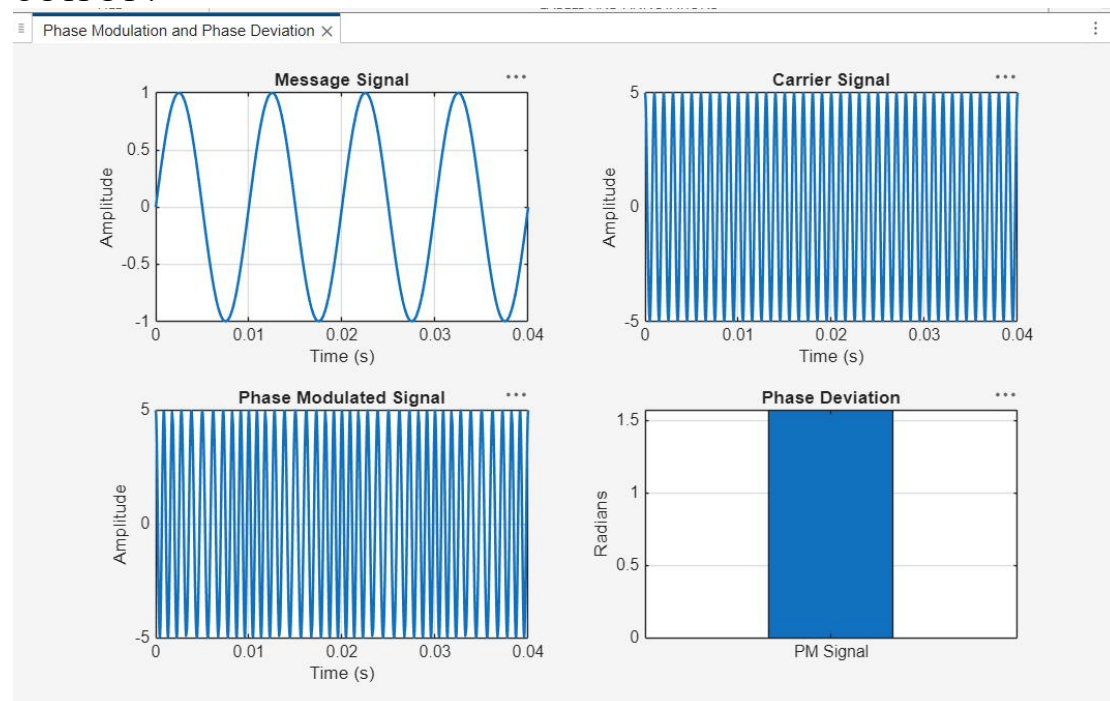
```
clear;
close all;
% Parameters
Am = 1;      % Message Amplitude (V)
fm = 100;    % Message Frequency (Hz)
Ac = 5;      % Carrier Amplitude (V)
fc = 1000;   % Carrier Frequency (Hz)
fs = 20*fc;  % Sampling Frequency (Hz)
kp = pi/2;   % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
phase_dev = kp*Am;
% PM Modulation Index
mp = phase_dev;
% Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Carrier Amplitude = %.2f V\n',Ac);
fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('Phase Deviation = %.2f radians\n',phase_dev);
fprintf('PM Modulation Index = %.2f\n',mp);
% Plotting
figure('Name','Phase Modulation and Phase Deviation',...
'NumberTitle','off');
subplot(2,2,1);
```

```

plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,c,'LineWidth',1.5);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(t,pm,'LineWidth',1.5);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,4);
bar(phase_dev);
title('Phase Deviation');
ylabel('Radians');
set(gca,'XTickLabel',{'PM Signal'});
grid on;

```

OUTPUT :



RESULTS : This program is successfully executed using the matlab .

3.2. POWER SPECTRAL DENSITY (PSD) ANALYSIS OF PHASE MODULATED (PM) SIGNALS USING MATLAB.

AIM : To generate and study power spectral density (psd) analysis of phase modulated (pm) signals using MATLAB .

SOFTWARE REQUIRED : MATLAB

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To compute the frequency spectrum of the PM signal using the Fast Fourier Transform (FFT).

CODE :

```
clc;

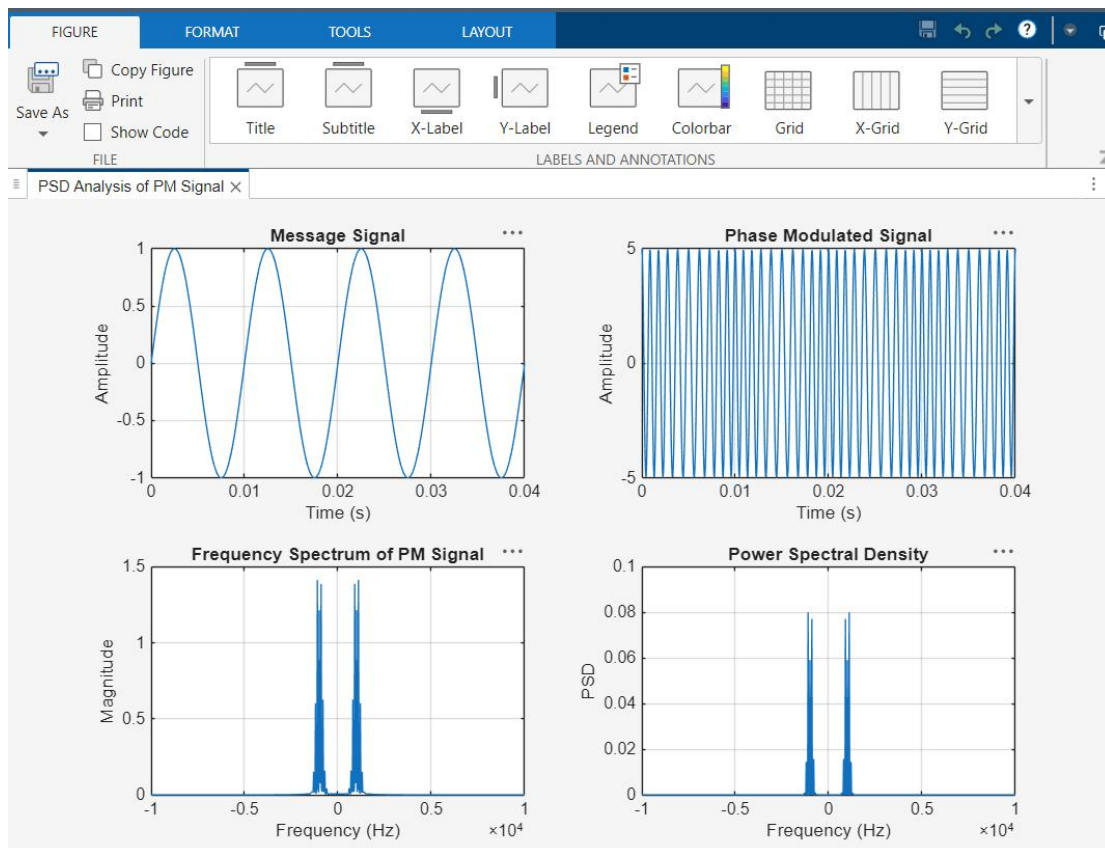
clear;
close all;
% Parameters
Am = 1;      % Message Amplitude (V)
fm = 100;    % Message Frequency (Hz)
Ac = 5;      % Carrier Amplitude (V)
fc = 1000;   % Carrier Frequency (Hz)
fs = 20*fc;  % Sampling Frequency (Hz)
kp = pi/2;   % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% PM Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
beta = kp*Am;
% FFT
N = length(pm);
f = linspace(-fs/2,fs/2,N);
PM_FFT = abs(fftshift(fft(pm)))/N;
% Power Spectral Density
PSD = (abs(fftshift(fft(pm))).^2)/(N*fs);
% Approximate Bandwidth
BW = 2*(beta+1)*fm;
fprintf('Phase Deviation = %.2f radians\n',beta);
fprintf('Estimated Bandwidth = %.2f Hz\n',BW);
% Plotting
figure('Name','PSD Analysis of PM Signal','NumberTitle','off');
subplot(2,2,1);
plot(t,m);
```

```

title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(f,PM_FFT);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(2,2,4);
plot(f,PSD);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD');
grid on;

```

OUTPUT :



RESULTS : This program is successfully executed using the matlab .